

Web Technology



Javascript

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Introduction

- **Most Popular Programming Language**
- **Webpages – Dynamic and Interactive**
- **Javascript?**

Introduction

- **Most Popular Programming Language**
- **Webpages – Dynamic and Interactive**
- **Javascript**
 - **Java**
 - **Indonesia – Java Coffee Bean**
 - **Javascript is different from Java**
 - **Javascript**
 - **Borrowed Concepts from Java**
 - **Object Oriented Programming**
 - **Java was popular (steal some popularity)**
 - **Netscape/Mosaic browser supported Java Applets**

Introduction

- **Most Popular Programming Language**
- **Webpages – Dynamic and Interactive**
- **Javascript**
- **Language?**

Introduction

Language

	English	Javascript	HTML
Alphabet	A-Z, a-z	A-Z, a-z	A-Z, a-z
Numbers	0-9	0-9, 0 and 1, 0-7, 0-F	0-9
Words	Words	Tokens	Nodes
Sentences	Grammar + Words = Sentences	Syntax + Tokens = Statements	Syntax + Nodes = Elements
Paragraph	Paragraph	Block	HTMLCollecti on
Chapter/Book	Chapter/Book	Program	Web Page
Library	Library	Library	Website

Introduction

- **Javascript – History**
 - **Reading Assignment - JavaScript – Wikipedia**

Introduction

- **Javascript – History**
 - **Reading Assignment - JavaScript – Wikipedia**
- **High-Level Language**

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- **Javascript – History**
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- **High-Level Language**
- **European Computer Manufacturer's Association (ECMA) script Standard**

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- **Javascript – History**
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- **High-Level Language**
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- **Dynamically Typed**

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- **Object Oriented Programming Language**

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- **Dynamically Typed**
- **Object Oriented Programming Language**
- **Interpreted/ Just-in-time Compiled**

Introduction

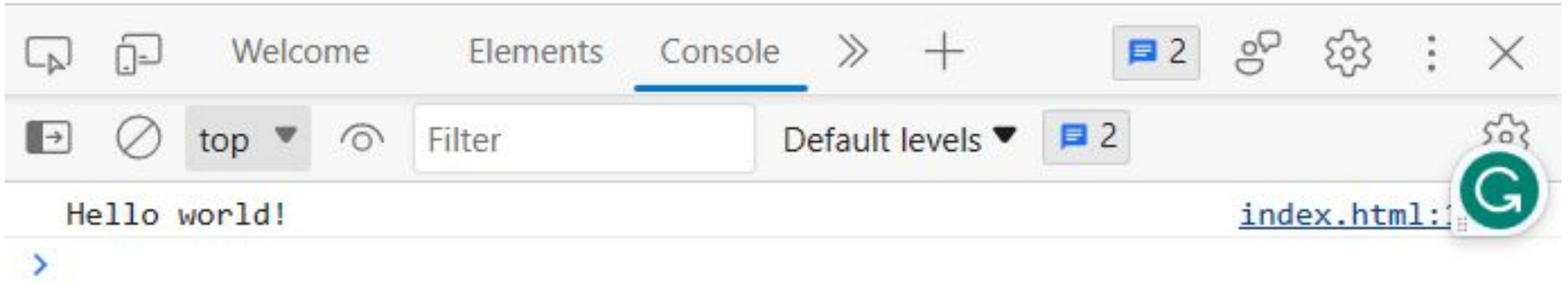
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- **Types of Programming?**

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- **Types of Programming**
 - **Inline Programming**
 - **External Programming**
- **<script> tag**

Introduction – Example 1

- The Hello World Program - Inline Programming
- `<script>` element
- Print in Console (Syntax: `console.log (“”);`)



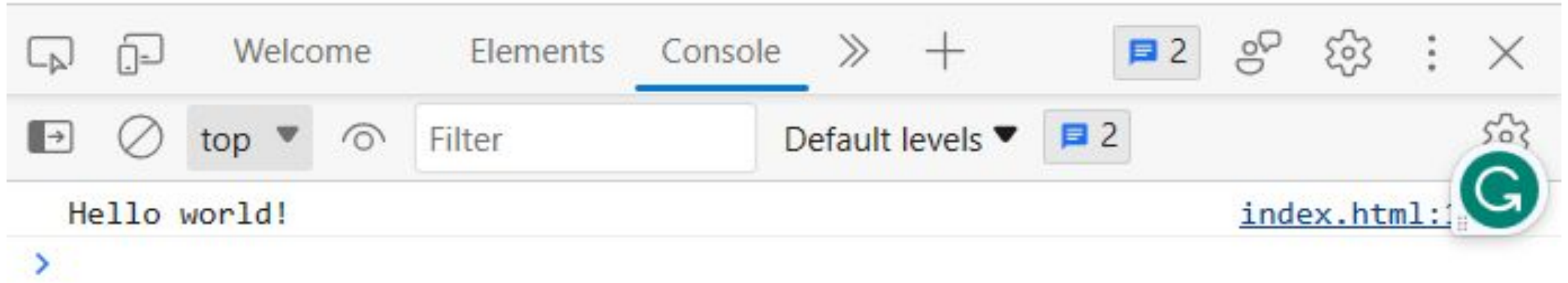
Introduction – Example 1

- **The Hello World Program - Inline Programming**

```
<!DOCTYPE html>
<html>
  <head>
    <title> Inline Javascript </title>
  </head>
  <body>
    <script>
      console.log("Hello world!");
    </script>
  </body>
</html>
```

Introduction – Example 2

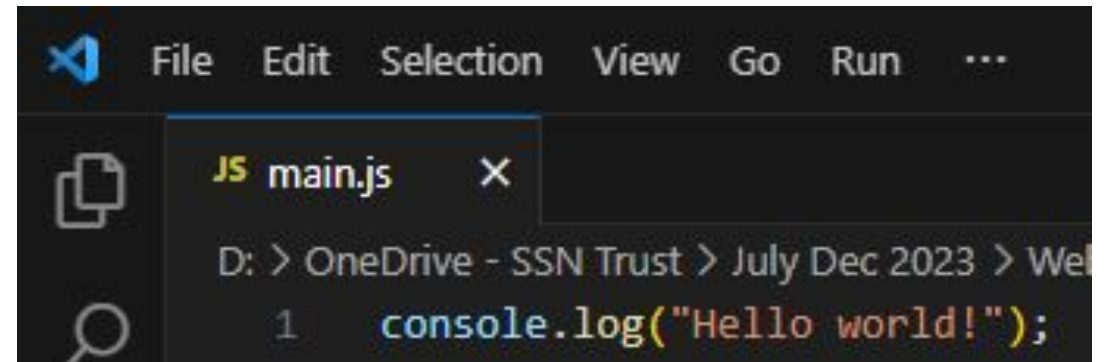
- The Hello World Program - External Programming (save as .js)
- `<script>` element. Include the Javascript file in src attribute of `<script>` element.
- Print in Console (Syntax: `console.log (“”);`)



Introduction – Example 2

- The Hello World Program - External Programming
- Script content - empty

```
<!DOCTYPE html>
<html>
  <head>
    <title> External Javascript </title>
  </head>
  <body>
    <script src="main.js">
    </script>
  </body>
</html>
```



Introduction – Example 3

- The Hello World Program - External Programming (save as .js)
- Pop an alert box. (Syntax: `alert(“”);`)

This page says

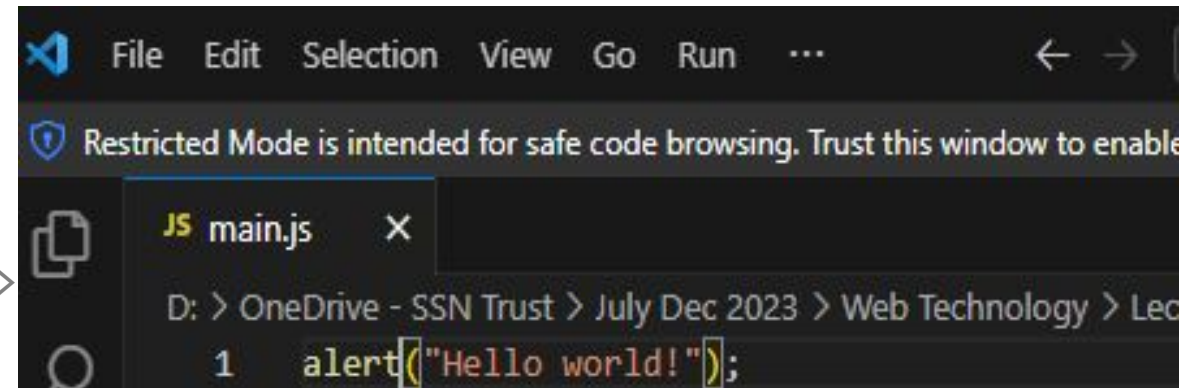
Hello world!

OK

Introduction – Example 3

- The Hello World Program - External Programming
- Script content - empty

```
<!DOCTYPE html>
<html>
  <head>
    <title> External Javascript </title>
  </head>
  <body>
    <script src="main.js">
    </script>
  </body>
</html>
```



Introduction – Example 4

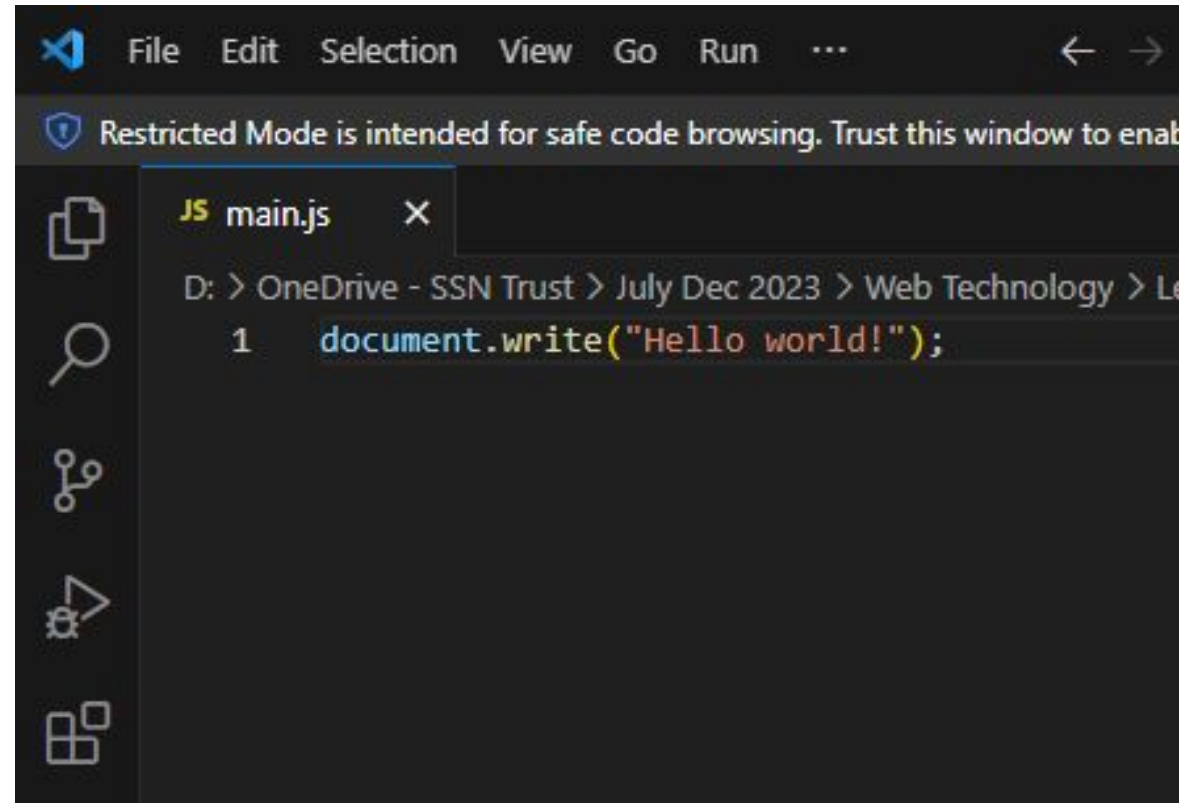
- **The Hello World Program - External Programming (save as .js)**
- **Pop an alert box. (Syntax: `document.write(“”);`)**



Introduction – Example 4

- The Hello World Program - External Programming
- Script content - empty

```
<!DOCTYPE html>
<html>
  <head>
    <title> External Javascript
  </title>
  </head>
  <body>
    <script src="main.js">
    </script>
  </body>
</html>
```



Best Practices

- 1. Use HTML tags with the same case (preferably lowercase).**
- 2. Filename to be based on functionality.**
- 3. When opening a tag, have a "tab space" for each line until you close the tag. It helps in the readability of your HTML document.**
- 4. Give a comment describing a web page as the first line of your HTML document.**
- 5. Keep HTML, CSS, and Javascript files separate**

Summary

- **Javascript Statistics**
- **Java vs Javascript**
- **Language**
- **Introduction**
- **Types of Programming**
- **Inline – Hello World - Inline**
- **External – Hello World – Inline**
- **External – Hello World - Alert**
- **Keywords**
- **Best Practices**

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**

THANK YOU

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